Project Proposal

Numerical Simulation of Manufacturing Processes

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April 20, 2026

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Triply Periodic Minimal Surface (TPMS)

TMPS are surfaces with the following mathematical properties [Wik25]:

- Nearly all studied TPMS are free of self-intersections (i.e. embedded in $\mathbb{R}^3$).
- All connected TPMS have genus ≥ 3, and in every lattice there exist orientable embedded TPMS of every genus ≥ 3.
- Embedded TPMS are orientable and divide space into two disjoint sub-volumes (labyrinths).

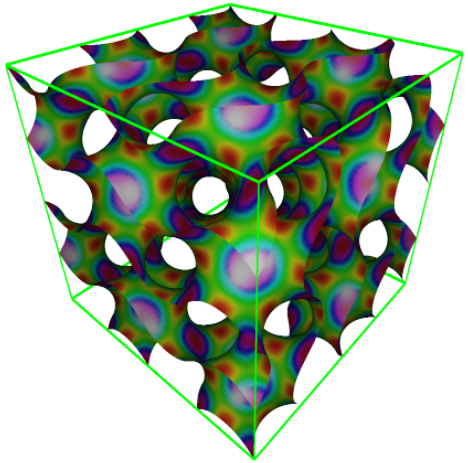
The mathematical background of TPMS is not particularly relevant for this project, but we will make use of the last property.

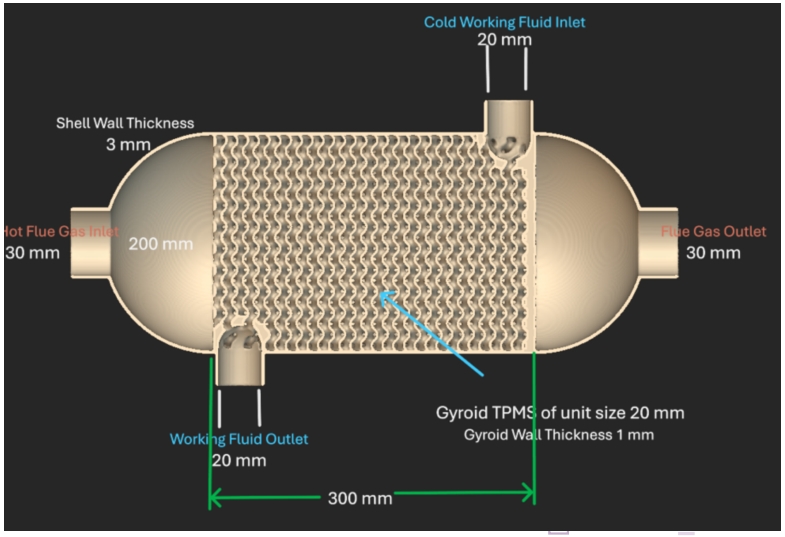
Gyroid-TPMS

Discovered by NASA scientist Alan Schoen in 1970, proved to be embedded by Grosse-Brauckmann and Wohlgemuth in 1996 [Wik26]. It is the structure that we will be treating in this project.

- Embedded; divides space into two *disjoint* sub-volumes.
- Can be approximated a the relatively simple equation

$$\sin x \cos y + \sin y \cos z + \sin z \cos x = 0$$





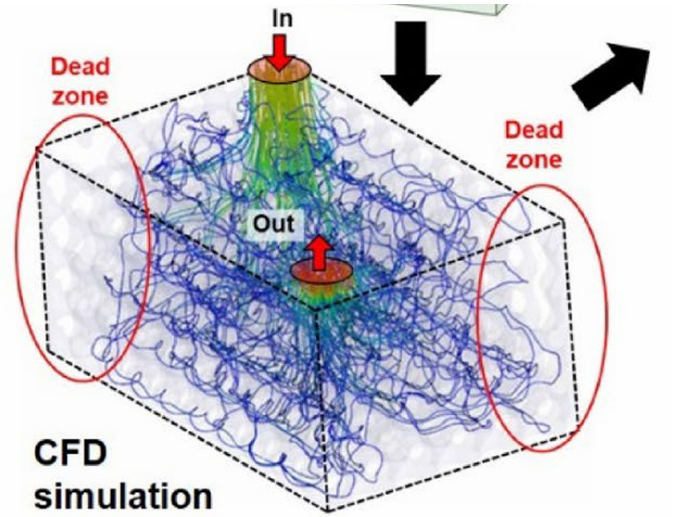
Advantages

TPMS provides a compelling alternative to traditional fin-based heat exchangers.

- High surface-to-volume ratio improves heat exchange.
- Continous TPMS walls generally provide good structural integrity (high stiffness).
- Geometry can be fine-tuned by function-based parameters.

The Problem

Dead zones appear in the structure. This causes poor flow utilization, which negatively affects the performance of the heat exchanger.



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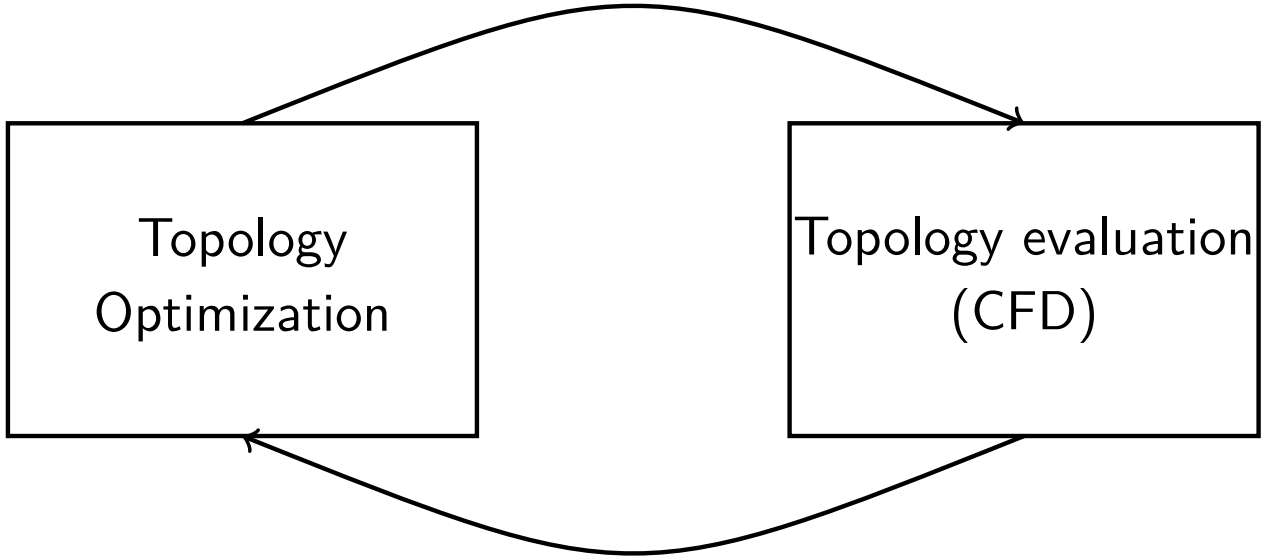
The Solution: Topology Optimization



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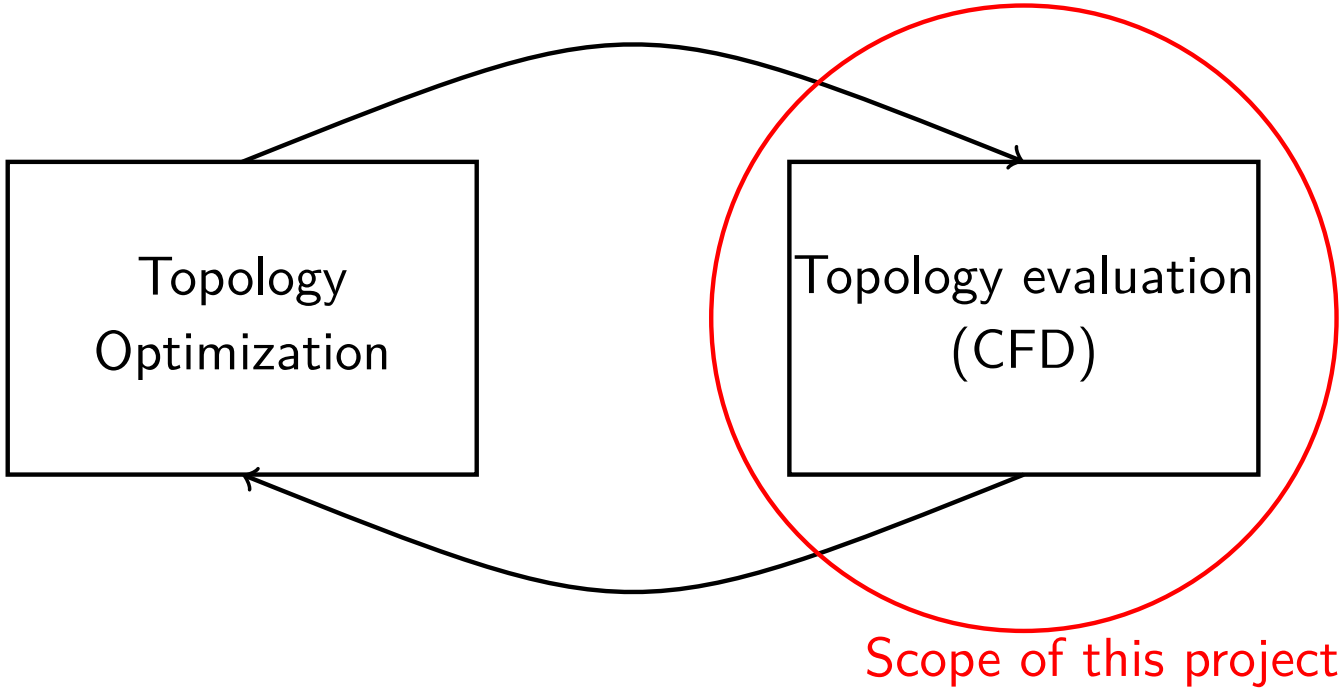

Current Methods

There are several ways to evaluate the performance of a given heat exchanger geometry.

- *Body-fitted Mesh* is the 'traditional' method which involves meshing of the structure, and evaluation in a FVM framework.
  - Well documented but computationally expensive for solving and re-meshing.

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  - Well documented but computationally expensive for solving and re-meshing.
- *Equivalent porous medium* replaces the TMPS with averaged porous material.
  - Very efficient but relatively poor performance [PBE25]
- *Brinkman penalization* is the method of choice as it can provide sufficient precision in combination with relatively fast results [YJW+23].

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How it works

- The solid regions are treated as porous, with permeability $\eta \to 0$ (approaching solid).
- A penalty term proportional to the Brinkman force is then added to the Navier-Stokes equation.
- The penalty term is chosen in such a way that the momentum of the fluids approaches zero inside the solid region, making it effectively non-permeable.

Why it works

- 'Traditional' CFD methods require geometry-specific meshes to be generated. This process can be computationally expensive, especially for complex (read TPMS) geometries.
- During topology optimization, the need to re-mesh the geometry at every iteration renders 'traditional' methods practically unusable for TPMS geometries.
- Methods based on Brinkman penalization mitigate these drawbacks by using only one mesh independent on the specific geometry of the solid region.
- This comes at the cost of lower accuracy, which is acceptable in the early design stage.

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How a Heat Exchanger Works

A heat exchanger transfers thermal energy between two fluid streams that are kept physically separated.

- A *hot* fluid and a *cold* fluid flow through adjacent channels, separated by a solid.
- Heat conducts through the solid wall; convection then distributes it into the bulk of each fluid stream.

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- The *energy* in both hot and cold fluids is modeled using:
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  - Diffusion
- The heat transfer across the solid wall is governed by conduction.

Heat Transfer Mechanisms

Heat is transferred between the two fluid streams via three mechanisms:

- *Conduction* diffuses heat through the solid wall:

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- *Diffusion* within the fluid also occurs, analogous to the process in the solid:

$$\mathbf{q}_{\text{diff}} = -k_f \nabla T$$

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- No viscous dissipation.
- No thermal radiation; only convection and conduction.
- No thermo-mechanical coupling (no stress/deformation).

The Equations

This leads us to the following equations to solve:

$$\nabla \cdot u = 0 \qquad \text{(Cont. eqn.)}$$

$$(u \cdot \nabla)u - \nabla \cdot \frac{1}{Re}\nabla^2 u + \nabla p = -\alpha u \qquad \text{(Momentum eqn.)}$$

$$u \cdot \nabla T - \nabla \cdot \frac{1}{Pe}\nabla T = 0 \qquad \text{(Energy eqn.)}$$

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Project Objective

Develop a numerical solver for steady-state heat transfer in a gyroid-based heat exchanger using Comsol.

- Solve the coupled flow and heat transfer equations in 3D using the Brinkman penalization framework.

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- Solve the coupled flow and heat transfer equations in 3D using the Brinkman penalization framework.
- The gyroid geometry is encoded implicitly through spatially varying penalization and material coefficients.
- No re-meshing is required when the geometry is modified, consistent with the Brinkman penalization philosophy.

Why Comsol?

- Comsol natively supports coupled PDE systems, allowing flow and heat transport to be solved simultaneously within one framework.

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- The Brinkman penalization term can be implemented as a user-defined volume force in the *Laminar Flow* module.
- Spatially varying material properties can be defined via analytic expressions, directly using the gyroid function.
- The same mesh is reused across different geometries, making the solver well suited for integration into a topology optimization loop.

Potential Challenges

- *Numerical stiffness* —the sharp transition in material properties at the fluid-solid interface may cause convergence difficulties and a smooth regularization may be needed.

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- *Mesh resolution* —resolving boundary layers near the gyroid wall in 3D is computationally demanding.
- *Penalization accuracy* —choosing the penalization parameter large enough to enforce no-slip without causing ill-conditioning is a known challenge in Brinkman methods.
- *Coupling convergence* —the nonlinear coupling between flow and temperature fields may require careful solver tuning in Comsol.

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